Homework – Penetration Scores: Law of Large Numbers and Central Limit Theorem

Name: _____ Course: _____ Date: _____

The simulation program is attached as `hmwk-penetration-lln-clt.html` (open it in any browser). All numbers below come from runs of its code (m = 2000 attackers, one run per model; a new run gives slightly different values).

1. Model

There are n systems and m attackers. Each attacker probes the n systems in sequence and obtains a score X_i on system i. The scores $X_1, X_2, \ldots$ are independent and identically distributed (i.i.d.). Two models are used:

- **Bernoulli**: $X_i \in \{0, 1\}$, $P(X_i = 1) = p$ (1 = a successful penetration). Here $\mu = E[X_i] = p$ and $\sigma^2 = \mathrm{Var}(X_i) = p(1 - p)$. In the simulation $p = 0.3$, so $\mu = 0.3$ and $\sigma^2 = 0.21$.
- **Rademacher**: $X_i \in \{-1, +1\}$, each with probability 1/2 (+1 = gain, −1 = loss). Here $\mu = 0$ and $\sigma^2 = 1$: the symmetric, zero-mean case.

For each attacker we study three quantities: the total score $S_n = X_1 + \ldots + X_n$, the average score S_n/n, and the normalized total score $S_n/\sqrt{n}$. The *m attackers* are m independent samples of these random variables, so histograms over the attackers estimate their distributions.

2. Theory

By independence, $E[S_n] = n\mu$ and $\mathrm{Var}(S_n) = n\sigma^2$. Scaling a random variable by c multiplies its variance by c^2, hence:

Quantity	Mean	Variance	Behavior as n grows
S_n	$n\mu$	$n\sigma^2$	spreads out (std. deviation $\sqrt{n}$)
S_n/n	μ	σ^2/n	variance → 0: concentrates at μ
$S_n/\sqrt{n}$	$\sqrt{n}\cdot\mu$	σ^2	variance constant; mean diverges unless $\mu = 0$

Law of Large Numbers (weak). For every $\varepsilon > 0$, $P(|S_n/n - \mu| > \varepsilon) \to 0$. This follows from Chebyshev's inequality: $P(|S_n/n - \mu| > \varepsilon) \leq \mathrm{Var}(S_n/n)/\varepsilon^2 = \sigma^2/(n\varepsilon^2)$. This is convergence in probability.

Central Limit Theorem. $(S_n - n\mu)/(\sigma\sqrt{n})$ converges in distribution to N(0,1). In the Rademacher case $\mu = 0$ and $\sigma = 1$, so this is exactly $S_n/\sqrt{n} \to N(0,1)$. In the Bernoulli case the normalized score $S_n/\sqrt{n}$ alone has mean $\sqrt{n}\cdot p$ that diverges; one must center it first.

3. Simulation results

Each cell shows the empirical value (theoretical value in brackets).

Bernoulli, p = 0.3 ($\mu = 0.3$, $\sigma^2 = 0.21$), $\varepsilon = 0.05$

n	Var(S_n)	Var(S_n/n)	Var($S_n/\sqrt{n}$)	P($\lvert S_n/n - \mu\rvert > \varepsilon$) (Chebyshev bound)	KS
10	2.03 (2.1)	0.0203 (0.021)	0.203 (0.21)	0.719 (⩽ 1)	0.151
100	19.6 (21)	0.00196 (0.0021)	0.196 (0.21)	0.211 (⩽ 0.84)	0.058
1000	209 (210)	0.00021 (0.00021)	0.209 (0.21)	0.001 (⩽ 0.084)	0.034
10000	2220 (2100)	0.00002 (0.000021)	0.222 (0.21)	0 (⩽ 0.008)	0.026

The empirical means were 3.0, 30.0, 300.5, 2998 for S_n (theory 3, 30, 300, 3000), about 0.30 for S_n/n at every n, and 0.95, 3.00, 9.50, 29.98 for $S_n/\sqrt{n}$ (theory 0.95, 3.0, 9.5, 30): the mean of $S_n/\sqrt{n}$ grows like $\sqrt{n}$.

Rademacher ($\mu = 0$, $\sigma^2 = 1$), $\varepsilon = 0.05$

n	Var(S_n)	Var(S_n/n)	Var($S_n/\sqrt{n}$)	P($\lvert S_n/n\rvert > \varepsilon$) (Chebyshev bound)	KS
10	9.92 (10)	0.0992 (0.1)	0.992 (1)	0.753 (⩽ 1)	0.130
100	98.6 (100)	0.00986 (0.01)	0.986 (1)	0.625 (⩽ 1)	0.047
1000	1037 (1000)	0.00104 (0.001)	1.037 (1)	0.109 (⩽ 0.40)	0.025
10000	10150 (10000)	0.0001 (0.0001)	1.015 (1)	0 (⩽ 0.04)	0.016

KS is the Kolmogorov–Smirnov distance between the empirical distribution of the standardized score $(S_n - n\mu)/(\sigma\sqrt{n})$ and the N(0,1) distribution. For m = 2000 samples, values below about $1.36/\sqrt{m} \approx 0.030$ are compatible with a normal distribution at the 5% level.

4. Discussion

LLN. For both models the variance of S_n/n shrinks like $1/n$, matching σ^2/n, and the fraction of attackers whose average differs from μ by more than ε falls toward zero (for Bernoulli: 0.72, 0.21, 0.001, 0). The empirical probabilities stay below Chebyshev's bound, which is correct but conservative (it uses only the variance). In the trajectory plots, S_k/k becomes a nearly horizontal line at μ (at 0 for Rademacher).

Variance under the three normalizations. $\mathrm{Var}(S_n)$ grows linearly with n; $\mathrm{Var}(S_n/n)$ decreases as 1/n; only $\mathrm{Var}(S_n/\sqrt{n})$ stays constant (σ^2), which is why $\sqrt{n}$ is the "right" scale to see a non-degenerate limit. The table confirms this in both models.

CLT. In the Rademacher case the histogram of $S_n/\sqrt{n}$ is jagged for n = 10 (the score takes only a few discrete values) and follows the N(0,1) density closely by n = 1000. The KS distance decreases from 0.13 to 0.016, below the 0.030 threshold from n = 1000. The same happens for the centered, standardized Bernoulli score, but more slowly (KS 0.034 at n = 1000, 0.026 at n = 10000) because the Bernoulli distribution with p = 0.3 is skewed, and the discrete (lattice) structure of S_n also contributes an error of order $1/\sqrt{n}$.

5. Interpretation: n systems and m attackers

- **System size n.** A larger n means each attacker's *total* score grows (linearly in the Bernoulli model) and becomes more variable in absolute terms (std. dev. $\sqrt{n}$), but the *average score per system* becomes more predictable (relative spread $1/\sqrt{n}$). With many systems, outcomes are determined by the success probability p, and luck matters less. For the zero-mean Rademacher model gains and losses cancel: the total is a random walk of typical size $\sqrt{n}$, so the net aggregate stays close to zero compared to n.
- **Number of attackers m.** Attackers are independent samples: increasing m does not change what a single attacker obtains, but it makes the histograms and the estimates of mean, variance and tail probabilities more accurate (their error decreases like $1/\sqrt{m}$). It also gives the defender a more reliable view of how many attackers exceed a given score.
- **Practical meaning.** For a defender, the LLN says that the long-run success rate of an attacker is p, so reducing p (hardening systems) directly lowers the aggregate score; the CLT quantifies the risk around that value: the number of compromised systems is approximately normal with mean np and standard deviation $\sqrt{(np(1-p))}$, which can be used to estimate the probability that more than a given number of systems are penetrated.